MIL-H-36343B (DSA-DM)
10 November 1966
Superseding
MIL-H-36343A (DM)
22 December 1964

MILITARY SPECIFICATION

HYDROMETERS, GRADUATED SCALE WITH THERMOMETER, API GRAVITY

1. SCOPE

1.1 Scope. This specification covers glass graduated-scale hydrometers, with integral thermometers, suitable for determining the American Petroleum Institute (API) gravity of petroleum products and other liquids having a surface tension not exceeding 33 dynes per centimeter.

1.2 Classification. API hydrometers shall be of the following types, classes, styles and ASTM hydrometer numbers (see 6.2):

Type I. API Gravity Thermohydrometers, Thermometer scale in body

ASTM Hydrometer Number	
51H	56H
52H	57H
53H	58H
54H	59H
55H	60H

Class 1. ASTM Suffix L Thermometer Range zero to 150° F.
Class 2. ASTM Suffix M Thermometer Range 30 to 180° F.
Class 3. ASTM Suffix H Thermometer Range 60 to 220° F.

Type II. API Gravity Thermohydrometers; Thermometer scale in body

Class 1. Thermometer Range zero to 110° F.
Class 2. Thermometer Range 20° to 130° F.

Style	Nominal API Gravity Range
1	- 1° to +11°
2	+ 9° to +21°
3	+ 19° to +31°
4	+ 29° to +41°
5	+ 39° to +51°
6	+ 49° to +61°
7	+ 59° to +71°
8	+ 69° to +81°

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM Designation	E100-62	ASTM Hydrometers
ASTM Designation	E126-62	Inspection, Test, and Standardization of Hydrometers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania)

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3. REQUIREMENTS

3.1 Materials. The materials used in the construction of type I and type II hydrometers shall be in accordance with ASTM Designation E100.

3.2 Construction.

3.2.1 Type I. Type I hydrometers shall be constructed in accordance with ASTM Designation E100.

3.2.2 Type II. Type II hydrometers shall consist of a glass stem and glass plummet, with an integral thermometer. The plummet end shall be filled with ballast as specified in ASTM E100. The thermometer scale shall be located in the body. The scales, thermometers, and ballast shall be held securely in place within the hydrometer. The thermometer bulb shall be embedded in a suitable heat-transfer medium enclosed within the body of the hydrometer.

3.3 Dimensions.

3.3.1 Type I. The dimensions of type I hydrometers shall be in accordance with ASTM Designation E100.

3.3.2 Type II. The overall length of type II hydrometers shall be 190 $\pm$ 5 millimeters (mm) and the bulb diameter shall be 14 $\pm$ 2 mm.

3.4 Thermometers. Class 1, class 2 and class 3 thermometers of type I hydrometer shall be in accordance with ASTM Designation E100. Class 1 and class 2 thermometers of type II hydrometers shall be graduated in 2° F. increments with major interval lines at 10° F. and with every 20° F. interval numbered. The error in Fahrenheit (F.) temperature scale for type I and type II shall not be greater than 1° F. when tested as specified in 4.4.1.

3.5 API Scale

3.5.1 Type 1. Shall be in accordance with ASTM Designation E 100.

3.5.2 Type II. The gravity scale shall be graduated in 0.2° API increments, with major interval lines and numbers every 1° API. The error shall be not greater than 0.2° API when tested as specified in 4.4.1.

3.6 Identification marking. The following information shall be legibly and permanently marked in a readily visible location on the inside of the hydrometer: Units of hydrometer scale, standard temperature (60° F.), ASTM Hydrometer No. for type I, manufacturer's serial number and the manufactuere's name or registered trademark.

3.7 Workmanship. Workmanship shall be first class throughout. The finished hydrometers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective
For visual examination		
Major defects	II	2.5
Minor defects	II	2.5
For dimensional examination	S-1	4.0

4.2.2 For test. Sampling for accuracy test shall be conducted in accordance with MIL-STD-105 for inspection level S-1. Acceptance number shall be zero.

4.3 Examination. The hydrometers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major
- 101 Ballast not secure in hydrometer.
- 102 Scale not legible, not correct, or not securely fixed in hydrometer.
- 103 Hydrometer not dry inside.
- 104 Glass not transparent or not smooth.
- 105 Stem and body not symmetrical to the vertical axis.
- 106 Thermometers not securely affixed to hydrometers.

Minor
- 201 Markings not correct, legible, or permanent.
- 202 Hydrometers not free from dirt, grease or other foreign material.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The hydrometers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The hydrometers shall be tested to determine the accuracy in accordance with ASTM Designation E126.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One hydrometer, of the class, size, and type specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall be in accordance with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-626, type CF, class domestic (method I closure shall be utilized).

5.2.3 Level C. The commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. In addition, the ASTM number, name and range shall be shown on the end of the carton (see 6.2).

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5.3.2 Interior packages. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend: "GLASS - HANDLE WITH CARE."

6. NOTES

6.1 Intended use. Hydrometers covered by this specification are intended to determine the API gravity ranges of liquid petroleum products.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Specify type, ASTM number (for type I only), class and style required (see 1.2).
(c) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: N

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-H-36343B (DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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